Implementing a Container in about 100 Lines of Go

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github.com/mhenke1/containerdemo

Background and Disclaimer

- Based on a blog post by Julian Friedman (Doctor Julz), 2016
  https://www.infoq.com/articles/build-a-container-golang/
- ex. IBM Bluemix (aka IBM Cloud Foundry); presented in the Böblingen lab around that time
- Adapted for a conference talk by Liz Rice, 2018
  https://www.youtube.com/watch?v=8fi7uSYlOdc
  https://github.com/lizrice/containers-from-scratch
- Adapted by me to cgroup v2
- Used in my Cloud Computing lecture

Goals

- Re-implement a simplified version of docker run in Go
- Start a container process
- Gradually implement some isolation (namespaces, chroot, mounts)
- Implement some resource limitation (cgroups v2)
- Command line:

```
// Docker original
docker run <image> <cmd> <params>

// Our version
go run main.go run <cmd> <params>
```

Make you envious that you did not invent Docker and get all the money

Isolation: Let's Get Our Hands Dirty

Live coding, one step at a time:

1. New hostname (UTS namespace)
2. New process IDs (PID namespace)
3. Own root filesystem (chroot)
4. Own /proc (mount) and a private mount namespace

Namespaces

Namespaces define what a process can see

- UTS: hostname (CLONE_NEWUTS)
- PID: process IDs (CLONE_NEWPID)
- Mount: mount points (CLONE_NEWNS)
- Network: devices, ports, routes (CLONE_NEWNET)
- User: user and group IDs (CLONE_NEWUSER)
- IPC: interprocess communication (CLONE_NEWIPC)
- Cgroup: view of the cgroup root (CLONE_NEWCGROUP)

The demo implements UTS, PID and Mount. See man 7 namespaces and clone(2).

Resource Limits: Let's Get Our Hands Dirty

Live coding with cgroups v2:

1. Create a cgroup and limit the number of processes (pids.max)
2. Run a fork bomb inside the container
3. Watch pids.current stay at the limit

Cgroups

Cgroups define what a process can use

- Controlled through a filesystem interface: /sys/fs/cgroup
- Memory: memory.max
- CPU: cpu.max, cpu.weight
- I/O: io.max
- Number of processes: pids.max (used in the demo)

See man 7 cgroups

Resources

- GitHub repo: https://github.com/mhenke1/containerdemo
- Julian Friedman: https://www.infoq.com/articles/build-a-container-golang/
- Liz Rice, talk: https://www.youtube.com/watch?v=8fi7uSYlOdc
- Liz Rice, code: https://github.com/lizrice/containers-from-scratch
- Linux man pages: namespaces(7), cgroups(7), clone(2)

Questions

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